

Graph Theory [L-T-P:3-1-0 Credit: 4] MAxxxxx

Prerequisite: NA

Category: New

Syllabus

1. Graphs, simple graphs, complete graphs, degree of vertices, subgraphs, adjacency matrix, incidence matrix, Isomorphism of graphs, self-complementary graphs, walks, trails, paths, cycles, connected graphs, Distances in graphs, eccentricity, radius, diameter. **[6 Lectures]**
2. Bipartite graphs, characterization of bipartite graphs, Eulerian graphs, characterization of Eulerian graphs, Hamiltonian graphs, Dirac's theorem. **[5 Lectures]**
3. Degree sequence, Graphic sequence, Havel-Hakimi theorem, Directed graphs, connectedness of directed graphs, tournaments, Digraphic sequences. **[5 Lectures]**
4. Trees, characterization of trees, spanning trees, Cayley's formula, Prufer codes, matrix-tree theorem (statement only), weighted graphs, minimal spanning trees, Kruskal's algorithm, Prim's algorithm, Dijkstra's shortest path algorithm, rooted trees, k-ary trees. **[5 Lectures]**
5. Matchings, maximal and maximum matchings, perfect matching, Hall's theorem (statement only), vertex covering number, edge covering number, independence number, relation among these parameters . **[5 Lectures]**
6. Vertex and edge connectivity of graphs, Whitney's theorem, Mengers Theorem(statement only) **[4 Lectures]**
7. Planar graphs, Euler's formula, consequences of Euler's formula, Kuratowski's Theorem(statement only), crossing number . **[5 Lectures]**
8. Vertex and edge colorings, chromatic number, edge chromatic number, Greedy coloring, bound on chromatic number, Brook's theorem, Vizing's theorem (statement only), Ramsey number. **[5 Lectures]**

Around 12 tutorial classes will be taken.

Reference Books:

1. Introduction to Graph Theory, by Douglas B West, Prentice Hall of India.
2. A first course in Graph Theory, by Gary Chartrand and Ping Zhang, Tata McGraw-Hill.
3. Graph Theory with Applications, by J.A. Bondy and U.S.R. Murty
4. Graphs and Digraphs, by Gary Chartrand

Teachers: Prof. Pratima Panigrahi and Dr. Swarup Kumar Panda